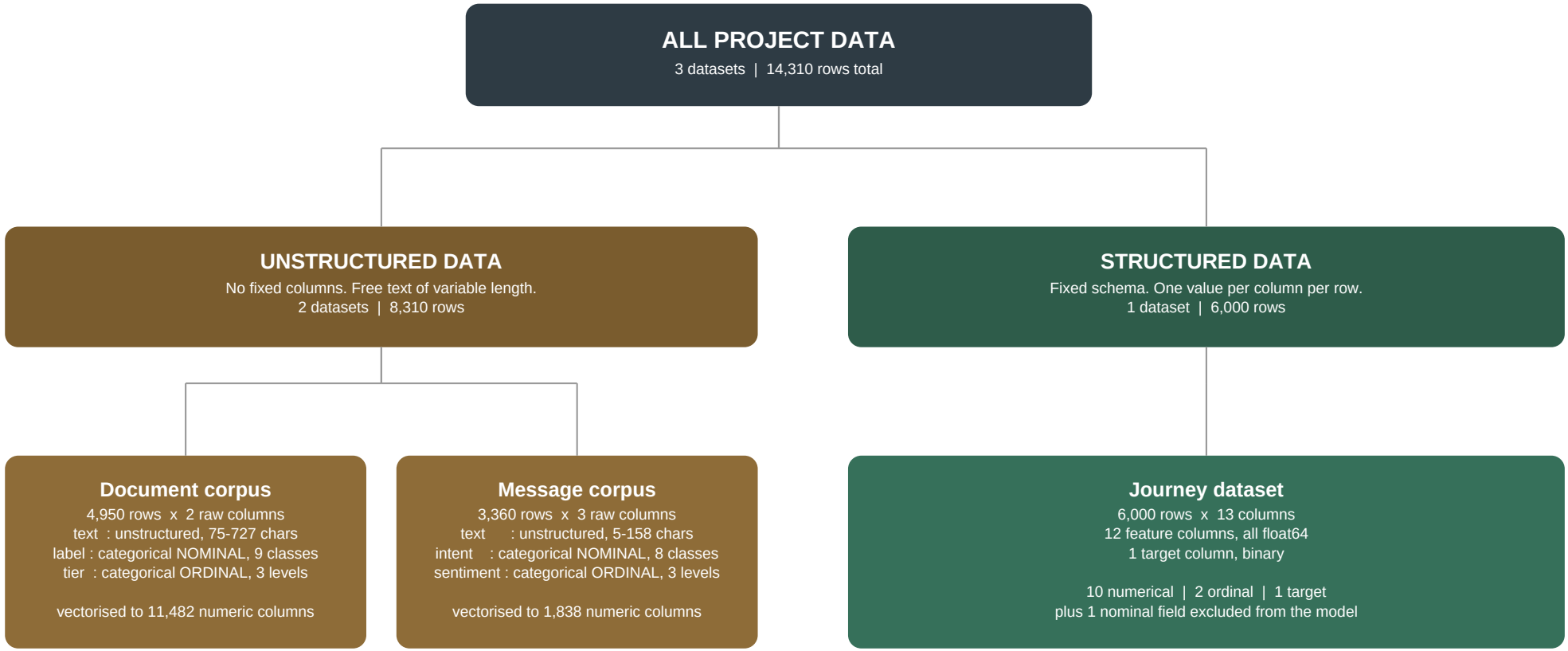
# Data Taxonomy: how the three datasets break down

Three datasets, 14,310 rows in total. Classification measured from the generators, not assumed.



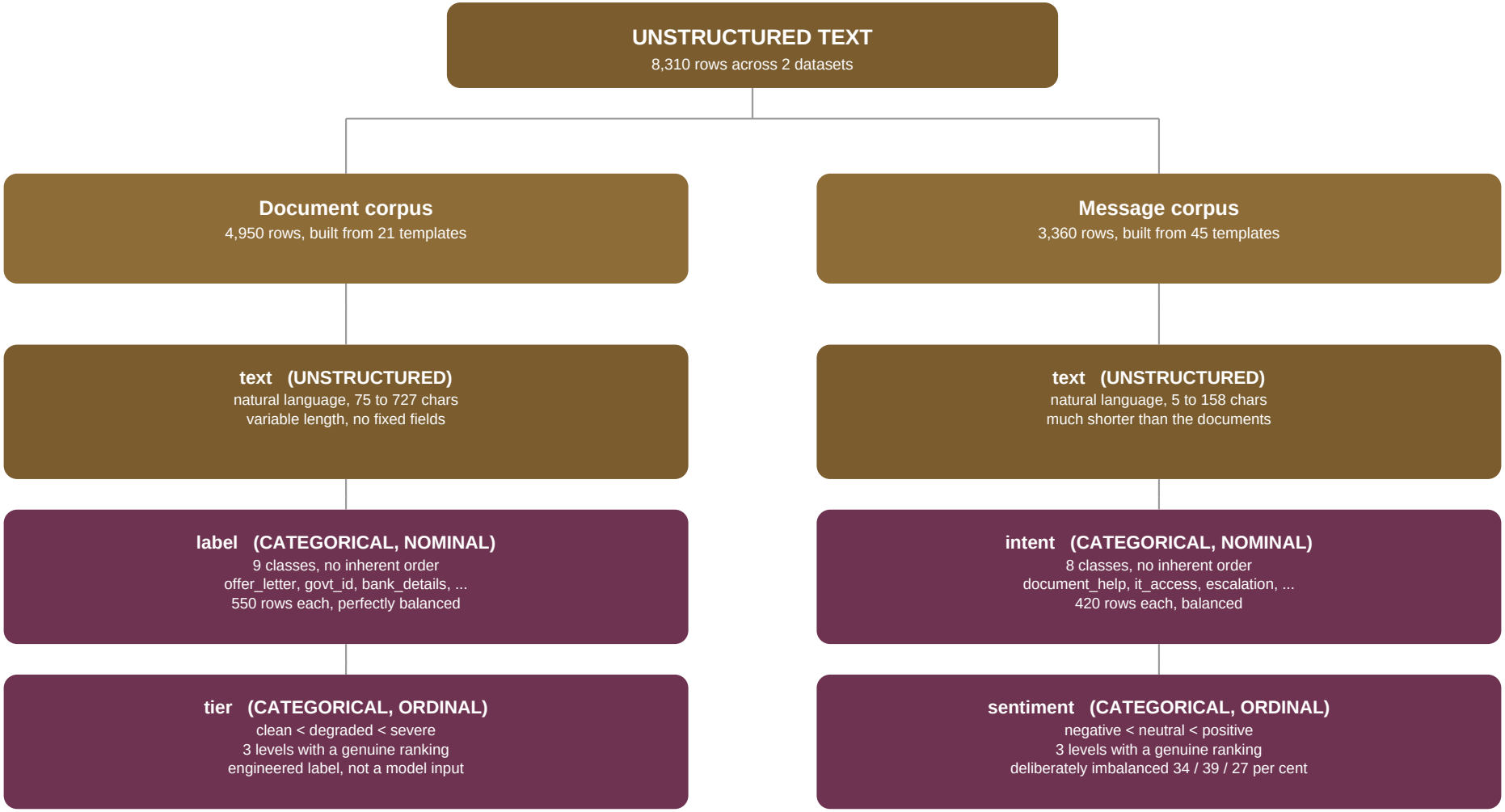

### Why the distinction matters

Unstructured data has no columns until it is vectorised. Asking how many columns a text dataset has has two correct answers, 2 or 3 raw fields, and thousands of engineered numeric features afterwards. The structured dataset is the only one where a column exists before any processing.

Both text datasets carry their labels as separate categorical fields, so each row is a text value plus one or two class labels.

# Unstructured data: the two text corpora

Each row is free text plus categorical labels. Numeric columns only exist after vectorisation.




### After vectorisation

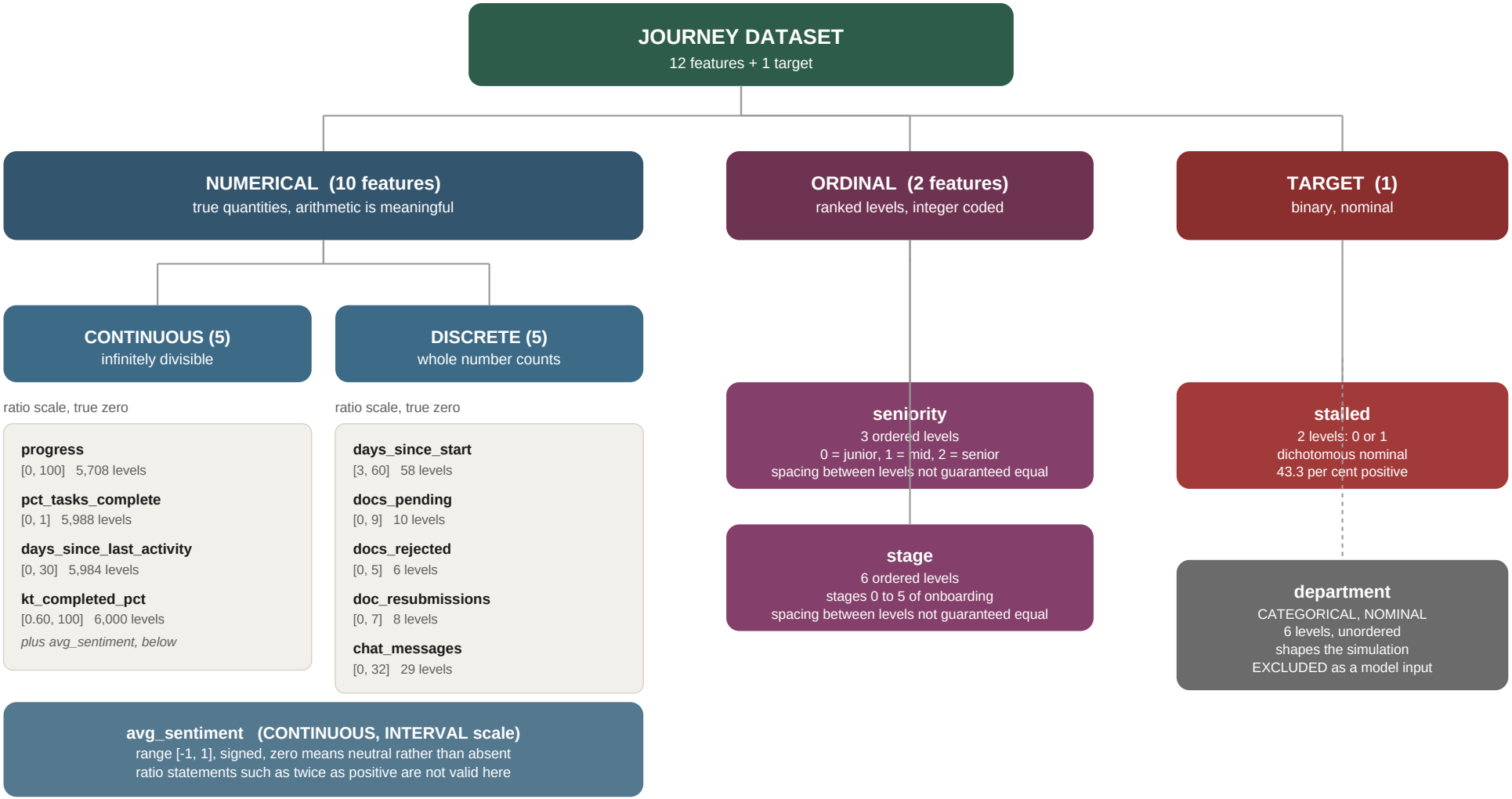
Documents become a 4,950 x 11,482 sparse matrix, 99.26 per cent zeros. Messages become 3,360 x 1,838, 98.78 per cent zeros.

Every engineered column is numerical and continuous, holding a TF-IDF weight between 0 and 1. The messages produce far fewer columns because they are much shorter, mean 73 characters against 348 for the documents.

The sentiment labels are additionally converted to a continuous score of p(positive) minus p(negative), bounded between -1 and +1, which is what the risk model consumes.

# Structured data: the journey dataset

6,000 rows by 13 columns. Every feature stored as float64, but the measurement scales differ.




### Two classification decisions worth defending

stage and seniority are ordinal but stored as numbers and treated as numeric by the model. That assumes the gap between stage 1 and 2 equals the gap between 4 and 5, which is not guaranteed. An alternative is one hot encoding, at the cost of losing the ordering. The current choice keeps the ordering and accepts the spacing assumption. department is nominal with no order and is deliberately not a model input. Predicting higher risk because of somebody's department invites disparate impact.

# Every variable in the project, classified

20 variables across the three datasets, including engineered and excluded ones.

| Dataset | Variable | Structure | Type | Scale | Levels / range | Role |
|---|---|---|---|---|---|---|
| Documents | text | Unstructured | Text | n/a | 75-727 chars | Input |
| Documents | label | Structured | Categorical | Nominal | 9 classes, 550 each | Target |
| Documents | tier | Structured | Categorical | Ordinal | clean < degraded < severe | Stratifier, not an input |
| Documents | tfidf_1..11482 | Engineered | Numerical | Ratio | 0 to 1, 99.26 pct sparse | Input after vectorising |
| Messages | text | Unstructured | Text | n/a | 5-158 chars | Input |
| Messages | intent | Structured | Categorical | Nominal | 8 classes, 420 each | Target |
| Messages | sentiment | Structured | Categorical | Ordinal | negative < neutral < positive | Target |
| Messages | sentiment_score | Engineered | Numerical | Interval | -1 to +1, continuous | Feature for risk model |
| Messages | tfidf_1..1838 | Engineered | Numerical | Ratio | 0 to 1, 98.78 pct sparse | Input after vectorising |
| Journeys | days_since_start | Structured | Numerical disc. | Ratio | [3, 60], 58 values | Model feature |
| Journeys | stage | Structured | Categorical | Ordinal | 6 ordered levels | Model feature |
| Journeys | progress | Structured | Numerical cont. | Ratio | [0, 100] continuous | Model feature |
| Journeys | pct_tasks_complete | Structured | Numerical cont. | Ratio | [0, 1] continuous | Model feature |
| Journeys | days_since_last_activity | Structured | Numerical cont. | Ratio | [0, 30] continuous | Model feature |
| Journeys | docs_pending | Structured | Numerical disc. | Ratio | [0, 9], 10 values | Model feature |
| Journeys | docs_rejected | Structured | Numerical disc. | Ratio | [0, 5], 6 values | Model feature |
| Journeys | doc_resubmissions | Structured | Numerical disc. | Ratio | [0, 7], 8 values | Model feature |
| Journeys | kt_completed_pct | Structured | Numerical cont. | Ratio | [0.60, 100] continuous | Model feature |
| Journeys | chat_messages | Structured | Numerical disc. | Ratio | [0, 32], 29 values | Model feature |
| Journeys | avg_sentiment | Structured | Numerical cont. | Interval | [-1, 1] continuous | Model feature |
| Journeys | seniority | Structured | Categorical | Ordinal | 3 ordered levels | Model feature |
| Journeys | stalled | Structured | Categorical | Nominal | 2 levels, 43.3 pct positive | Target |
| Journeys | department | Structured | Categorical | Nominal | 6 levels | Excluded from the model |

# Definitions used, and why each classification was chosen

The terms applied on the preceding pages, with the project's own examples.

| | |
|---|---|
| **Structured** | Fixed schema. Every row has the same columns and one value per column. Only the journey dataset qualifies before processing. |
| **Unstructured** | No fixed fields. Free text of variable length. Both text corpora qualify, and neither has columns until vectorised. |
| **Categorical** | Takes one of a fixed set of labels. Arithmetic on the codes is meaningless. Document label, intent, sentiment, stalled, department. |
| **Nominal** | Categorical with no inherent order. Nothing makes govt_id greater than bank_details. Applies to document label, intent, stalled, department. |
| **Ordinal** | Categorical with a genuine ranking but no guaranteed equal spacing. clean < degraded < severe, negative < neutral < positive, junior < mid < senior, stage 0 through 5. |
| **Numerical, discrete** | Whole number counts. Cannot take a value between two adjacent levels. docs_pending, docs_rejected, doc_resubmissions, chat_messages, days_since_start. |
| **Numerical, continuous** | Any value within the range. progress, pct_tasks_complete, days_since_last_activity, kt_completed_pct, avg_sentiment. |
| **Ratio scale** | Has a true zero, so ratios are meaningful. Zero documents pending really is none, and four pending is twice two. |
| **Interval scale** | Differences are meaningful but zero is a reference point rather than an absence. avg_sentiment sits here: zero means neutral, not no sentiment, so twice as positive is not a valid statement. |
| **Binary or dichotomous** | Categorical with exactly two levels. The target, stalled, is 0 or 1 and positive in 43.3 per cent of rows. |
| **Engineered** | Not present in the raw data, produced by a transformation. The TF-IDF columns and the sentiment score. |
| **Clustered text** | Not used in this project. The text corpora are labelled, so classification is supervised. Clustering would group unlabelled documents by similarity, which is a possible next phase extension. |